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Engineer to Excel

TECHNICAL QUESTIONS FOR UST GLOBAL – SET2

1. Your task is to complete a function “count_heads()” that takes two inputs N and R. The function should return the probability of getting exactly R heads on N successive tosses of a fair coin. A fair coin has an equal probability of landing a head or a tail (i.e. 0.5) on each toss.

Example 1

Input: 1 1

Output: 0.500000

Example 2

Input: 4 3

Output: 0.250000

2. Write a program that will print the sum of diagonal elements of a 10X10 matrix. The program will take a total of 100 numbers as input (10 numbers will be input per line and each number will be separated by a space).

Example 1

Input: 1 2 3 4 5 6 7 8 9 0

0 1 2 3 4 5 6 7 8 0

3 4 5 6 7 8 9 6 4 0

2 3 4 5 6 7 8 9 3 2

3 4 5 6 7 4 3 2 1 3

3 4 5 6 2 4 4 2 4 6

2 3 4 6 2 4 6 2 3 5

2 3 5 6 2 4 6 2 3 5

2 4 6 2 1 4 3 3 5 2

3 3 5 2 4 6 2 1 4 6

Output: 42

Example 2

Input: 1 22 33 44 55 66 77 88 99 100

100 1 88 77 66 55 44 33 22 11

88 88 1 66 55 44 33 22 11 100

88 77 66 1 44 33 22 11 100 99

77 66 55 44 1 22 11 88 99 100

66 55 44 33 22 1 77 88 99 100

44 33 22 11 100 99 1 77 66 55

33 22 11 100 99 88 77 1 55 44

22 11 100 99 88 77 66 55 1 33

100 11 22 33 44 55 99 88 77 1

Output: 10

3. Write a program that will take one string as input. The program will then remove vowels a, e, i, o, and u (in lower or upper case) from the string. If there are two or more vowels that occur together then the program shall ignore all of those vowels.

Example 1

Input: Cat

Output: Ct

Example 2

Input: Compuuter

Output: Cmpuutr

4. Write a program that will take a string as input. The program will then determine whether each left parenthesis ‘(’ has a matching right parenthesis ‘)’ and also all the ‘)’ has a consecutive ‘(‘. If so, the program will print 0 else the program will print 1.

Example 1

Input: HELLO AND (WELCOME (TO THE) TCEA (CONTEST)TODAY)IS (SATURDAY())

Output: 0

Example 2

Input: (9*(7-2)*(1*5)

Output: 0

5. Write a program to find out and display prime numbers from the given list of integers. The program will accept input in two lines. First-line contains a number indicating the total number of integers in the list and the second line contains integers separated by spaces.

Example 1

Input: 5

4 6 9 3 7

Output: 3 7

Example 2

Input: 10

8 10 3 12 7 15 11 2 17 26

Output: 3 7 11 2 17

6. Write a program that will take a number as input. The program will convert the inputted number to a format, which will contain an integer part and a fraction part.

The fraction part needs to be reduced to its lowest representation. The input will not contain any repeating decimals e.g. 1.3333…33. The output will be :

The integer part of the number ++fraction using a ’/’

Example 1

Input: 2.95

Output: 2 19/20

Example 2

Input: 3.08

Output: 3 2/25